

Question number	Answer	Mark
3(a)(i)	A	(1)

Question number	Answer	Mark
3(a)(ii)	C	(1)

Question number	Answer	Mark
3(b)(i)	B at 0.4 seconds	(1)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	A calculation in which: • volume of blood per beat from graph = 59 cm³ (1) • heart rate calculated from graph = 75 bpm (1) • volume of blood converted into dm³ (1) Example of calculation: $(59 \times 75) \div 1000 = 4.425 \text{ dm}^3$	Accept 109 – 50 Correct answer with no working shown gains all three marks	(3)

Question number	Answer	Mark
3(c)	An explanation that includes the following points: • dissociation curve for the llama is to the left of that for the human (1) • therefore llama haemoglobin has a higher affinity for oxygen (1) • llama haemoglobin will be fully saturated with oxygen at lower partial pressures (1) • this is necessary as there is less oxygen available in the atmosphere at high altitudes where llamas live (1)	(4)